

Unit 8, Lesson 11: Surface Area of a Rectangular Pyramid

Benchmark

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

Learning Target

- ☐ I can find the surface area of a rectangular pyramid using a net or the figure.

8.11.1 Warm-Up: Would You Rather?

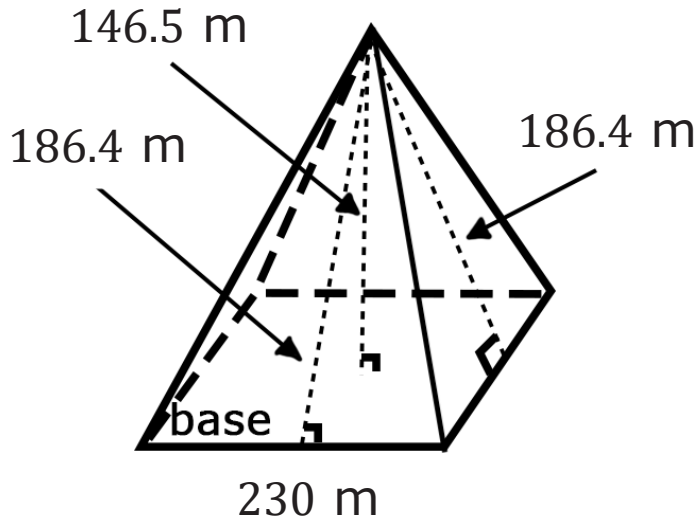
1. Would you rather have Room 1 or Room 2?

	Room 1	Room 2
Length	20 feet	14 feet
Perimeter	58 feet	56 feet

2. Write an explanation of your choice.

8.11.2 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Great Pyramid of Giza, also known as the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.



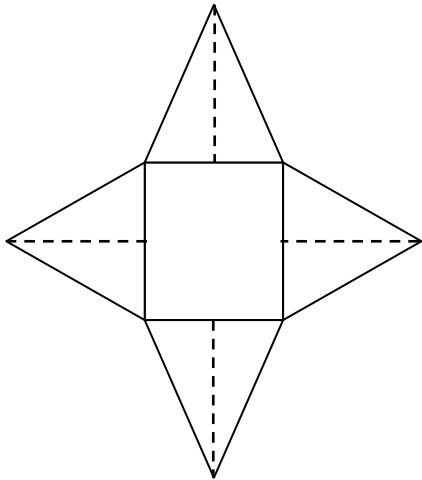
Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?

The slant height of the pyramid represents the height of each triangular face.



When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the **net**.



Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$ $A = \underline{\hspace{1cm}}$
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$ $A = \underline{\hspace{1cm}}$
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$ $A = \underline{\hspace{1cm}}$
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$ $A = \underline{\hspace{1cm}}$
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$ $A = \underline{\hspace{1cm}}$
		Surface Area:

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need

the sum of the

○ height

○ area

 of the base plus the area of

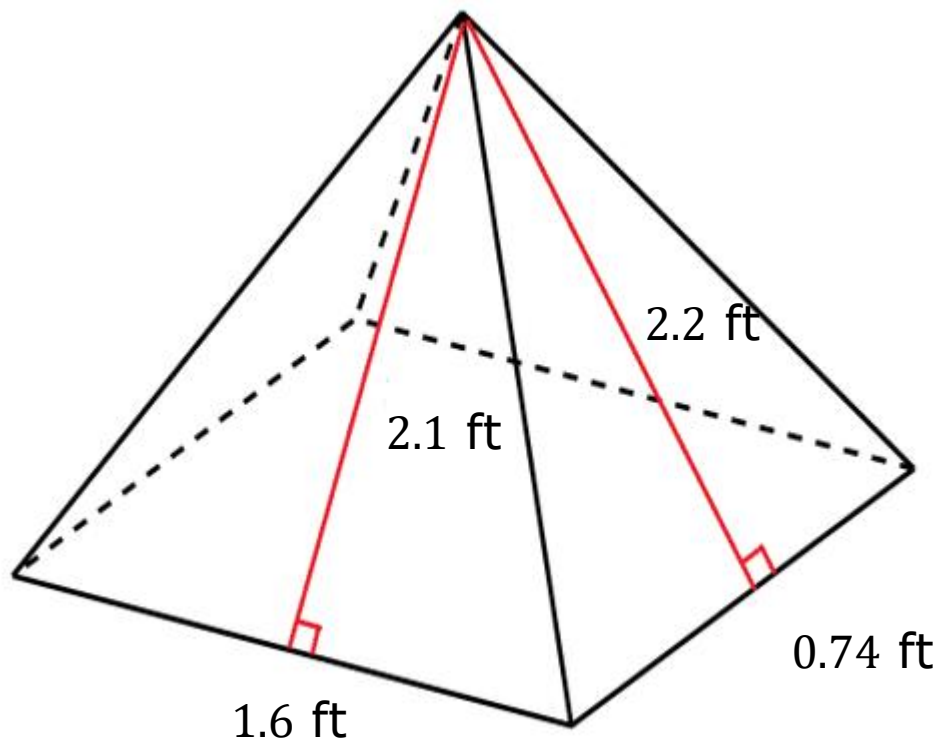
○ one of

○ all

 the other faces.

8.11.3 Collaboration: Surface Area of a Rectangular Pyramid

Papier-mache is a type of art that uses paper and glue. Jose is using the art of papier-mache to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.

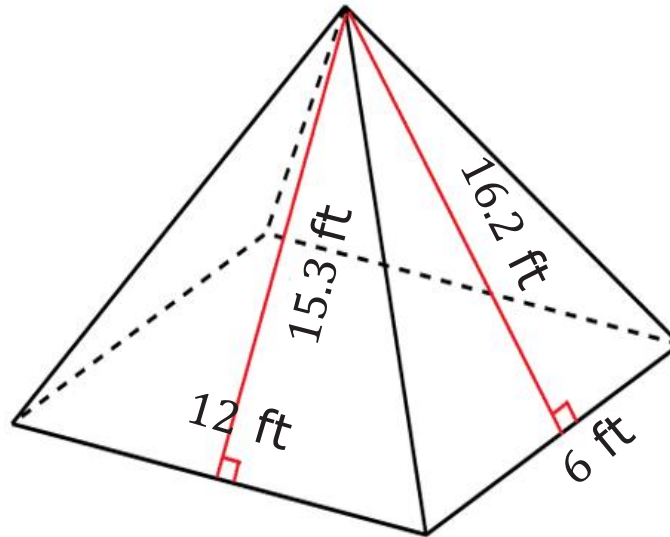


1. Work with your partner to find the surface area of the pyramid.

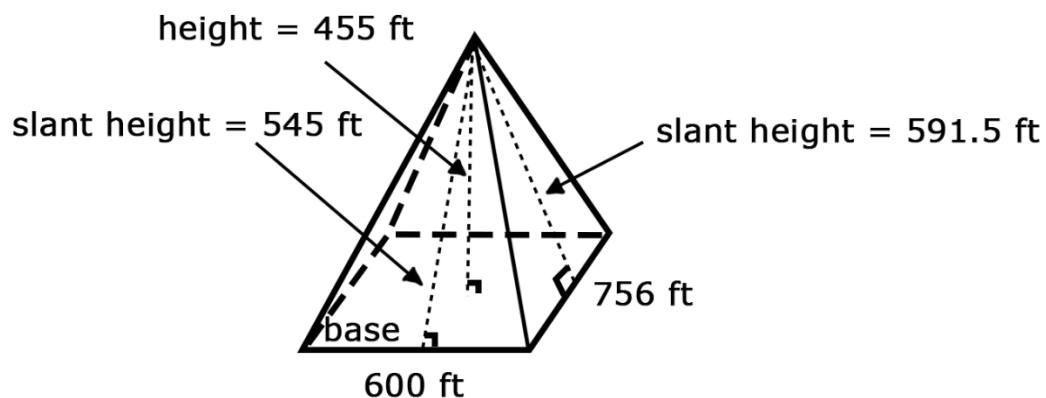
8.11.4 Your Turn!

Calculate the surface area for the pyramids.

1. Calculate the surface area of the pyramid.



2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.



3. At the entrance to the Louvre Museum in Paris, France, is an architectural pyramid that was completed in 1989. The pyramid has a square base with side lengths of 34 meters, and the pyramid stands at a height of 21.6 meters and a slant height of 27.5 meters. Using this information, calculate the surface area of the rectangular pyramid.



8.11.5 Wrap-Up: Ticket Out the Door

1. Find the surface area of a rectangular pyramid that has a square base with side lengths 75 centimeters (cm), a height of 100 cm, and a slant height of 125 cm.

